# Distance-hereditary graphs characterize minimum-fusion synthesis of graph states

**A machine-discovered result with a formally verified core**

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**Abstract**

In fusion-based quantum computation a graph state $|G\rangle$ on $N$ qubits is assembled from small entangled resource states by destructive two-qubit fusion measurements. We study the minimum number $\mathrm{F}(G)$ of fusions required in the model whose resource is the three-qubit Greenberger–Horne–Zeilinger state and whose fusion is the type-II Bell measurement (the $\mathrm{GHZ}_3$ model). Counting qubits and connected components gives the universal bound $\mathrm{F}(G) \geq N - 3$ for connected targets. We determine exactly which graphs attain it: $\mathrm{F}(G) = N - 3$ if and only if $G$ is *distance-hereditary*—equivalently, of rank-width at most one; equivalently, $C_5$-vertex-minor-free.

The result was produced by an autonomous formal-methods harness rather than by hand. A language model (Fable 5), permitted to emit only machine-checkable artifacts and never given execution access, classified all 585 local-complementation orbits of connected graphs on at most nine vertices by whether they attain the bound, every certificate accepted by two independent stabilizer engines; proposed the distance-hereditary characterization from the 44 labeled orbits on at most seven vertices, after which all 541 held-out orbits on eight and nine vertices proved consistent; and formalized the core of the result in Lean 4 against an adversarially hardened trust gate. Formalizing the forward direction (distance-hereditary $\Rightarrow \mathrm{F} = N - 3$) required extending the production model with a second fusion primitive, itself verified against both engines; the reverse direction is a Lean theorem at the level of that model, and an exact counting argument lifts it to the physical minimum. Under the harness's discipline no claim is recorded above the status of a heuristic without a machine artifact, and the mathematics, the conjecture, and the Lean proofs were authored by the model.

## 1 Introduction

Fusion-based quantum computation (FBQC) builds fault-tolerant computation from a supply of small, identical entangled *resource states* that are stitched together by *fusions*: destructive joint measurements of two qubits [1]. Because the resource states and the fusions are the physical cost drivers of a photonic architecture, a basic question for any target entangled state is how few fusions suffice to prepare it. For graph states—the stabilizer states in one-to-one correspondence with simple graphs [2]—this is a combinatorial optimization over the graph.

We work in a specific instance, which we call the $\mathrm{GHZ}_3$ model: the resource is the three-qubit GHZ state $|\mathrm{GHZ}_3\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$, whose graph is the three-vertex path, and each fusion is a type-II (destructive) Bell measurement. Since a stabilizer state is defined only up to single-qubit Clifford operations, the natural cost is a function of the local-Clifford orbit. Writing $N = |V(G)|$, we set

$$\mathrm{F}(G) \;=\; \text{the least number of fusions producing a state locally Clifford equivalent to } |G\rangle,$$

starting from a fresh supply of $|\mathrm{GHZ}_3\rangle$ states, with single-qubit Clifford operations free throughout.

A short accounting of qubits and connected components gives a universal lower bound $\mathrm{F}(G) \geq N-3$ (Section 2). The graphs meeting it form the extremal class of the problem, and our main result identifies that class exactly.

**Theorem 1** (Characterization)**.** *For a connected graph $G$ on $N \geq 3$ vertices in the* $\mathrm{GHZ}_3$ *model,*

$$\mathrm{F}(G) = N-3 \quad \Longleftrightarrow \quad G \textit{ is distance-hereditary.}$$

Distance-hereditary graphs [5, 6] are exactly the graphs of rank-width at most one [7], the graphs with no vertex-minor isomorphic to the five-cycle $C_5$, and the graphs obtainable from a single vertex by repeated addition of pendant vertices and twins. Rank-width is invariant under local complementation [7, 4], and F is invariant under local Clifford operations; the characterization thus matches the fusion cost to a local-Clifford-invariant measure of the entanglement structure of $|G\rangle$.

**Contributions.** The mathematical content of the paper is Theorem 1 and its constituent parts:

1. a certified classification of all 585 connected local-complementation orbits with $N \leq 9$ by whether $\mathrm{F} = N-3$, together with exact values strictly above the bound for thirteen orbits; every certificate carries a witness accepted by two independent stabilizer engines (Section 4);
2. Lean 4 proofs of the universal lower bound and of $\mathrm{F} = N-3$ for paths, stars, all trees, complete graphs $K_N$, and complete bipartite graphs $K_{m,n}$ (Section 6);
3. a Lean 4 proof of the forward direction of Theorem 1 (distance-hereditary $\Rightarrow$ $\mathrm{F} = N-3$), obtained by extending the production model with a second, engine-verified fusion primitive (Section 7); and
4. a Lean 4 equivalence identifying the graphs producible in the extended model at the floor with the distance-hereditary graphs, together with the counting argument that lifts its reverse direction to the physical minimum (Section 8).

The paper also documents the method. Every item above was discovered or written by an autonomous harness driving a single language model under an explicit epistemic discipline: no statement is recorded above the status HEURISTIC without an accompanying machine artifact, the model that does the mathematics never executes anything itself, and the formalization is checked by a trust gate that was hardened against adversarial submissions. We describe the harness in Section 3 because the provenance bears on how the results should be read.

**What is proved, and in what sense.** Because these distinctions matter for a machine-aided result, we state them plainly and keep to them throughout.

- *Lean theorem*: a statement checked by the Lean 4 kernel and a hardened gate, using only the axioms `propext`, `Classical.choice`, `Quot.sound`, against MATHLIB pinned at `v4.31.0`. The lower bound, the five families, the forward direction, and the model-level equivalence are of this kind.
- *Engine-verified*: a fact about the physical fusion operation confirmed by two independently implemented stabilizer simulators that share no code. The two fusion primitives used by the production model are of this kind; inside Lean they are graph rewrites, and their right to be called "one fusion" rests on the engine agreement rather than on a derivation of qubit semantics in Lean.

- *Prose argument*: the counting bridge of Section 8, which shows that a schedule attaining $f = N - 3$ has a shape realizable in the extended model. It is proved here at the ordinary standard of rigor of a mathematics paper but is not formalized; the remark in Section 8 explains why.

Theorem 1 combines all three. Independently of its proof, the biconditional was checked against the certified tablebase on all 585 orbits and holds with no exception.

## 2 The $\mathrm{GHZ}_3$ fusion model and the lower bound

**Graph states and local complementation.** To a simple graph $G = (V, E)$ one associates the stabilizer state $|G\rangle = \prod_{\{u,v\}\in E} \mathrm{CZ}_{uv} |+\rangle^{\otimes V}$. Local Clifford equivalence of graph states is generated by *local complementation*: complementing at a vertex $v$ toggles the adjacency of every pair of neighbors of $v$ [3, 4]. We write $G \sim_{\mathrm{LC}} H$ for the resulting equivalence and note that both F and the rank-width of $G$ are constant on $\sim_{\mathrm{LC}}$ orbits.

**Resource and fusion.** The resource $|\mathrm{GHZ}_3\rangle$ has graph the three-vertex path $P_3$. A type-II fusion destructively measures the two-qubit Paulis $X \otimes Z$ and $Z \otimes X$ on a pair of qubits; on a graph state it removes the two measured vertices and rewires their neighborhoods, up to Pauli byproducts that are absorbed into the local-Clifford equivalence. Because intermediate and final states are tracked only up to local Cliffords, fusion conventions that differ by single-qubit Cliffords on the measured pair yield the same cost F.

**Model dependence.** The value F is specific to this resource and this fusion. Other natural models—for instance models with emitter-generated caterpillar resources—assign different values, and the extremal class can differ between models. We fix the $\mathrm{GHZ}_3$ model throughout and use externally tabulated values from other models only as soft cross-checks, never as ground truth.

**The universal lower bound.** Suppose a schedule consumes $g$ copies of $|\mathrm{GHZ}_3\rangle$ and performs $f$ fusions to produce a connected $N$-vertex graph state. Each resource contributes three qubits and each fusion consumes two, so

$$3g - 2f = N. \tag{1}$$

The $g$ resources begin as $g$ connected components and the target is a single component. Components merge only when a fusion consumes one qubit from each of two distinct components, and a resource that never merges into the final component must instead have all three of its qubits consumed by fusions, which costs more than merging it; an induction over the schedule gives $f \geq g - 1$. Substituting $g \leq f + 1$ into (1) yields $N = 3g - 2f \leq 3(f + 1) - 2f = f + 3$, hence

**Proposition 2** (Universal lower bound)**.** $\mathrm{F}(G) \geq N - 3$ *for every connected $G$ on $N$ vertices.*

This is the content of the Lean declaration `fusion_cost_lower_bound`. The component book-keeping is not assumed but derived from a merge model of the schedule; in particular the Lean proof covers the degenerate steps in which a fusion splits a component or consumes one outright, which the one-line argument above glosses over.

## 3 The Empiricist harness

The results were obtained by a harness ("Empiricist") that drives a language model, Fable 5, through research tasks while forbidding it any unverified step. Four of its design commitments bear on how the results should be read.

**An append-only epistemic ledger.** Every artifact—a computed value, a construction, a conjecture, a Lean module—enters a ledger at a status in the ordered set

$$\text{REFUTED} < \text{HEURISTIC} < \text{CONJECTURED} < \text{VERIFIED}_N < \text{CERTIFIED} < \text{FORMALIZED},$$

and rises only when accompanied by an evidence row from a verifier. REFUTED is terminal. Nothing the model merely asserts is recorded above HEURISTIC. Artifacts are addressed by content hash and certificates embed exact version pins, so results are reproducible from the ledger alone.

**The model never executes.** Fable 5 emits structured JSON—a conjecture, a construction, or a complete Lean module—and the harness runs every program in a sandbox. This keeps the boundary between a model's suggestion and a checked fact sharp, and it is what lets us attribute the mathematics to the model while still calling the results verified. The harness itself—ledger, executor, verifiers, gate—is conventional software.

**Two independent physical verifiers.** Fusion outcomes are checked by two stabilizer engines that share no code: a tableau simulator built on STIM [11] and a GF(2) bitmask implementation written independently from the physical specification. A fusion move is accepted only when both engines agree on the resulting stabilizer state up to local Clifford equivalence; disagreement is treated as a fault that halts the campaign. The closed-form graph rewrite for each resource merge is trusted only because it matches both engines on a fuzz test against random input states, cross-checked against NAUTY's canonical forms [12].

**A hardened Lean trust gate.** A submitted Lean 4 [9] module is elaborated in a sandbox with network and process creation denied and an ephemeral write jail, against a frozen read-only snapshot of a pinned MATHLIB [10]. The kernel result is re-checked by an independent checker; the axiom set is required to lie within {`propext`, `Classical.choice`, `Quot.sound`}; every import must resolve to a member of an explicit trusted set; and the build directory is swept for residue after elaboration. The gate was developed against successive rounds of adversarial red-teaming (poisoned olean files, time-of-check/time-of-use swaps, axiom-hiding elaborators), with each exploit repaired before the gate was used in production.

**The formalization loop.** To formalize a statement the model emits a complete Lean module as JSON; the gate either accepts it or returns structured diagnostics, including the exact proof obligations remaining at unfilled holes. The model revises across rounds until the module passes. The loop uses no shell and no interactive proof assistant on the model's side; the model develops a proof by reading goal states the gate reports, not by running Lean.

# 4 The certified tablebase

The harness organized the connected graphs with $3 \le N \le 9$ into local-complementation orbits (relabeling and local complementation identified); there are 585. For each orbit the ledger records whether $\mathrm{F} = N - 3$, with a machine certificate either way: when the orbit is extremal, a fusion schedule with $N - 3$ fusions whose outcome both engines accept, meeting the lower bound of Proposition 2; when it is not, a certified refutation of every schedule at the floor. (By the counting argument of Section 8, a schedule with $N - 3$ fusions consumes $N - 2$ resources and every one of its fusions joins two components, so the floor schedules form a finite space; each fusion outcome

| $N$ | 3 | 4 | 5 | 6 | 7 | 8 | 9 | total |
|---|---|---|---|---|---|---|---|---|
| connected orbits | 1 | 2 | 4 | 11 | 26 | 101 | 440 | 585 |
| extremal ($\mathrm{F} = N - 3$) | 1 | 2 | 3 | 8 | 15 | 42 | 104 | 175 |
| non-extremal ($\mathrm{F} > N - 3$) | 0 | 0 | 1 | 3 | 11 | 59 | 336 | 410 |

Table 1: Connected local-complementation orbits by vertex count, split by whether the orbit attains the universal minimum $\mathrm{F} = N - 3$. The extremal set has 175 orbits; all are distance-hereditary.

in the exhaustion is computed by both engines.) The resulting object—585 rows, each an orbit representative with its verdict and certificates—is a single content-addressed artifact at status VERIFIED$_N$.

Table 1 records the split. Two further outcomes of the campaign are worth noting. First, exact values strictly above the floor were closed for thirteen orbits. Ten sit one tier up, at $\mathrm{F} = N$ (one orbit at $N = 5$, two at $N = 6$, seven at $N = 7$), and were closed by deterministic search. Three sit two tiers up, at $\mathrm{F} = N+3$, and were closed by model-proposed schedules: a six-vertex three-regular orbit at $\mathrm{F} = 9$, realized with eight resources, nine fusions, and three local-complementation steps, and two seven-vertex orbits at $\mathrm{F} = 10$; each schedule was independently re-verified by both engines. Second, the natural refinement $\mathrm{F} = N-3+3(\mathrm{rw}(G)-1)$, which would make the fusion cost a linear function of rank-width, is false: a seven-vertex orbit of rank-width two carries a certified bound $\mathrm{F} \geq N + 3 = 10$, whereas the formula predicts $\mathrm{F} = N = 7$. Rank-width captures the extremal boundary exactly but does not determine the cost above it.

## 5 The extremal class is distance-hereditary

**Discovery.** The harness presented the model with the complete labeled catalogue of the 44 orbits with $N \leq 7$, each as an edge list tagged extremal or not, with no invariant, class name, or hint in the prompt beyond the true observation that the target property is a local-complementation invariant. The model proposed that the extremal orbits are exactly the distance-hereditary graphs, supplying the standard equivalent characterizations (metric, rank-width at most one, forbidden induced house/gem/domino/long-cycle, and pendant/twin reducibility), the relevant invariance results [5, 7, 4], and the identification of the smallest obstruction: the unique non-extremal five-vertex orbit is the local-complementation orbit of $C_5$.

**Grounding.** A deterministic checker—Bandelt–Mulder pendant/twin elimination, which reduces a graph to a single vertex if and only if it is distance-hereditary—was run on all 585 orbits and compared against the extremal label. The classification $\mathrm{F} = N - 3 \Leftrightarrow$ distance-hereditary holds with zero exceptions, including on the 541 held-out orbits with $N \in \{8, 9\}$ that the model never saw. The proposal was recorded as a CONJECTURED artifact, promoted from HEURISTIC by this grounding.

Every family for which we prove $\mathrm{F} = N - 3$ in Section 6 is distance-hereditary, which is why each attains the bound: trees, complete graphs, and complete bipartite graphs all have rank-width at most one.

# 6 Formalized families and the lower bound

The Lean development begins with Proposition 2, formalized as a self-contained module, and with a production predicate. $\mathsf{Prod}_k(G)$ holds when the concrete graph $G$ is obtained from $P_3$ by $k$ successive *leaf-merges*, up to isomorphism; a leaf-merge attaches one pendant vertex, and its graph rewrite is proved in Lean to be isomorphic to that of the physical resource merge at a leaf qubit, whose physical meaning is engine-verified (Section 3). The predicate deliberately has no local-complementation constructor, so it generates exactly the trees; local equivalence enters only when a statement about F is made.

**Theorem 3** (Families)**.** *In the* $\mathrm{GHZ}_3$ *model,* $\mathrm{F}(G) = N - 3$ *for* $G$ *a path, a star, any tree, a complete graph* $K_N$*, or a complete bipartite graph* $K_{m,n}$*. Each is a Lean theorem: the lower bound is Proposition 2, and the upper bound exhibits a graph* $H \sim_{\mathrm{LC}} G$ *with* $\mathsf{Prod}_{N-3}(H)$*.*

The proofs reduce each family to a producible tree up to local complementation: $K_N$ is one local complementation from a star, $K_{m,n}$ is three from a double star, and trees are produced directly by leaf-removal induction, of which paths and stars are special cases. The statements are faithful in a strict sense: each carries the general lower bound in its own statement—quantifying over arbitrary fusion schedules, not only the exhibited construction—so that a theorem cannot be true for a vacuous or circular reason.

*Remark* 4 (Induction must track concrete graphs)*.* Although every statement about F is invariant under local complementation, the inductions cannot be conducted on $\sim_{\mathrm{LC}}$ orbits, because pendant addition does not descend to orbits: $P_3 \sim_{\mathrm{LC}} K_3$, yet attaching a pendant to the center of $P_3$ yields the four-vertex star, while attaching a pendant to a vertex of $K_3$ yields the paw, which is LC-equivalent to $P_4$ and not to the star. The Lean development therefore carries exact graphs through every induction and applies local complementation only explicitly.

# 7 The forward direction of the characterization

Theorem 3 covers exactly the graphs locally equivalent to a tree: since $\mathsf{Prod}$ generates the trees and nothing else, it can witness the floor only on the local-complementation orbit of a tree. This is a strict subclass of the distance-hereditary graphs. A direct check on the tablebase shows that of the 175 extremal orbits, 93 contain a tree in their local-complementation orbit and 82 do not—two at $N = 6$, four at $N = 7$, nineteen at $N = 8$, and fifty-seven at $N = 9$. (The orbit enumerations behind these counts ran to completion; a safety cap of 200000 graphs per orbit was never reached.) The forward direction of Theorem 1 is therefore not provable in that model.

**The missing primitive.** The obstruction is a missing operation, not a difficult proof. The physical resource merge fuses one qubit $a$ of the current state with one qubit of a fresh $|\mathrm{GHZ}_3\rangle$, and up to the symmetry of $P_3$ there are two choices: the fused resource qubit is a leaf or the center. The leaf choice attaches a pendant. The center choice removes the fused state vertex $a$ and introduces the two resource leaves, each adjacent to exactly the former neighborhood of $a$ and not to each other; up to relabeling this is $\mathrm{addFalseTwin}(G, a)$. In Lean the center-merge rewrite is proved isomorphic to false-twin addition (`ghz3CenterMerge_iso_addFalseTwin`); physically, the closed form matches the engine outcome up to local Clifford equivalence on 400 of 400 random input states, with the two engines agreeing with each other on all 400—the same discipline already applied to the leaf-merge. The center merge is a genuine one-fusion primitive; the production model had simply never included it.

**The extended model.** We formalize the production predicate $\mathsf{Prod}_k^+$ generated by five constructors: the base $P_3$; the leaf-merge (pendant) and center-merge (false twin), each adding one vertex and one fusion; local complementation, which is free; and graph isomorphism. Because distance-hereditary graphs are closed under pendant addition, twin addition, local complementation, and isomorphism, every $\mathsf{Prod}^+$-producible graph is distance-hereditary. The converse is the content of the forward direction.

**Theorem 5** (Forward direction, formalized)**.** *Let* $\mathsf{DH}$ *denote the class generated from the connected three-vertex graphs ($P_3$ and $K_3$) by pendant, false-twin, and true-twin additions, closed under isomorphism; these are exactly the connected distance-hereditary graphs on at least three vertices [5]. Then, as a Lean theorem,*

$$G \in \mathsf{DH} \implies \mathsf{Prod}_{N-3}^+(G),$$

*and hence, with Proposition 2,* $\mathrm{F}(G) = N - 3$*.*

The proof is a structural induction on the Bandelt–Mulder build, carrying the exact graph as Remark 4 requires. A pendant addition maps to a leaf-merge; a false-twin addition maps to a center-merge; a true-twin addition is realized as local complementation, then a leaf-merge, then local complementation, using the graph identity

$$\mathrm{lc}\big(\mathrm{addPendant}(G, u),\, u\big) \;=\; \mathrm{addTrueTwin}\big(\mathrm{lc}(G, u),\, u\big)$$

together with the involutivity of local complementation. No search for a witness is needed: tracking the exact graph through the extended model turns each build step into one constructor. This is what makes the theorem short once the extended model is in place, and it is the step at which the "locally equivalent to a tree" obstruction dissolves.

## 8 The reverse direction

The reverse direction is proved at the level of the extended model and then lifted to the physical minimum by counting.

**Theorem 6** (Model characterization, formalized)**.** *As a Lean theorem,*

$$\mathsf{Prod}_{N-3}^+(G) \iff G \in \mathsf{DH}.$$

The right-to-left implication is Theorem 5. The left-to-right implication is a structural induction on $\mathsf{Prod}^+$: the base is a three-vertex core, the leaf-merge and center-merge map back to a pendant and a false twin through the same faithfulness isomorphisms used above, and local complementation and isomorphism map to themselves. Making the last two steps immediate requires recording that $\mathsf{DH}$ is closed under local complementation; since rank-width at most one is a local-complementation invariant this does not enlarge the class, and it replaces what would otherwise be a family of transport identities by a single constructor.

**From the model to the physical minimum.** Theorem 6 concerns the extended model; the passage to F is an exact counting argument. Suppose a schedule attains $f = \mathrm{F}(G) = N - 3$ using $g$ resources. By (1), $3g = N + 2f = 3N - 6$, so $g = N - 2$, and the component bound $f \geq g - 1 = N - 3$ holds with equality. Equality forces every fusion to consume one qubit from each of two distinct components: there are no intra-component fusions. Since a fusion destroys both of its qubits, the $N - 3$ fusions act on pairwise disjoint qubit pairs and commute; assigning to each fusion the pair of resources owning its qubits yields a connected graph on the $N-2$ resources with $N-3$ edges—a tree. Reordering the fusions along a root-first traversal of this tree gives an equivalent schedule in which every fusion merges one fresh resource into a single growing component. Up to the free local Cliffords, a fresh resource carries the graph $P_3$ or its local complement $K_3$, so each merge fuses the growing state with a leaf of $P_3$, the center of $P_3$, or a vertex of $K_3$; on graphs these realize a pendant, a false twin, or a true twin respectively, and all three are available in $\mathsf{Prod}^+$ (the true twin via the identity of Section 7), while local Cliffords on the growing state are local complementations, free in $\mathsf{Prod}^+$. The reordered schedule is therefore a $\mathsf{Prod}^+$ derivation. Hence $\mathrm{F}(G) = N-3$ implies $\mathsf{Prod}^+_{N-3}(G)$, and with Theorem 6, $G$ is distance-hereditary. Combined with Theorem 5 this proves Theorem 1.

*Remark* 7. The counting bridge is not internalized as a single Lean theorem, for a concrete reason: an intra-component fusion has no closed-form graph rewrite—the harness computes it only through the stabilizer engines—so a general-schedule model in which Lean could state "$\mathrm{F}(G) = N-3$" directly is not available. The alternative classical route to the reverse direction (every graph that is not distance-hereditary has $\mathrm{F} > N-3$) would run through rank-width lower bounds [7, 8], and rank-width is not present in MATHLIB. The counting argument above avoids both obstacles at the standard of rigor of this paper, at the price of leaving the bridge outside Lean.

## 9 Discussion

**The modeled boundary.** The two fusion primitives are the one place where the formal development rests on something other than the Lean kernel. Inside Lean they are graph rewrites; their right to be called "one fusion" comes from agreement between two independent stabilizer engines and a canonical-form cross-check, not from a derivation of qubit semantics in Lean. This is a deliberate boundary, stated rather than hidden: the combinatorial optimization is proved, and the physical primitive it optimizes over is pinned down by redundant simulation. A fully first-principles account would formalize the stabilizer formalism and the Bell measurement in Lean and re-derive the two rewrites; that is a separate and larger undertaking.

**Discovery under discipline.** The characterization was proposed by the model from labeled data and then survived an exhaustive deterministic check; the forward direction was formalized by the model against a gate it could not talk its way past. The discipline has a measurable payoff beyond trust: the deterministic recheck of the characterization corrected an arithmetic error in an earlier hand-written note (the extremal/non-extremal split is 175/410, not 185/400), and the faithfulness requirement on Theorem 3 caught a near-circular statement in which the construction class had been silently baked into the claim. A workflow that permits unchecked assertion would have surfaced neither.

**Open questions.** Three stand out. First, the physical reverse direction as a single Lean theorem, which appears to require either a formalized intra-component fusion rewrite or rank-width in MATHLIB. Second, a finer invariant governing F above the extremal boundary, where rank-width is insufficient (Section 4). Third, the complexity of computing F: the characterization makes the decision problem $\mathrm{F}(G) \stackrel{?}{=} N-3$ solvable in polynomial time, since distance-hereditary graphs are recognizable in polynomial time via pendant/twin elimination [5], but the complexity of computing F in general is open.

## 10 Conclusion

In the $GHZ_3$ fusion model the graphs that attain the universal minimum fusion count are precisely the distance-hereditary graphs. The characterization was proposed by a language model from certified data, holds with no exception on all 585 connected local-complementation orbits with at most nine vertices, and is formalized in Lean 4 in the forward direction and, in the reverse direction, at the level of a production model that an exact counting argument connects to the physical minimum. The development was carried out end to end by an autonomous harness under a discipline that records nothing above a heuristic without a machine artifact; that discipline is both how the result was obtained and the sense in which it should be trusted.

## A Principal Lean declarations

Table 2 lists the main formalized statements. All are checked by the hardened gate and use only the axioms {`propext`, `Classical.choice`, `Quot.sound`}. The foundation modules are committed and reusable; the theorem modules are recorded as gate-certified ledger artifacts.

| declaration | statement | role |
|---|---|---|
| `fusion_cost_lower_bound` | $\mathrm{F}(G) \geq N-3$ (schedule form) | Prop. 2 |
| `pathGraph_min_fusions` | $\mathrm{F}(\mathrm{path}_N) = N-3$ | Thm. 3 |
| `star_min_fusions` | $\mathrm{F}(\mathrm{star}_N) = N-3$ | Thm. 3 |
| `tree_min_fusions` | $\mathrm{F}(T) = N-3$ for all trees | Thm. 3 |
| `complete_min_fusions` | $\mathrm{F}(K_N) = N-3$ | Thm. 3 |
| `completeBipartite_min_fusions` | $\mathrm{F}(K_{m,n}) = N-3$ | Thm. 3 |
| `ghz3CenterMerge_iso_addFalseTwin` | center-merge = false twin | Sec. 7 |
| `dh_forward` / `dh_min_fusions` | $\mathsf{DH} \Rightarrow \mathsf{Prod}^{+}_{N-3}$ | Thm. 5 |
| `dh_characterization` | $\mathsf{Prod}^{+}_{N-3}(G) \Leftrightarrow G \in \mathsf{DH}$ | Thm. 6 |

Table 2: Principal formalized declarations, against MATHLIB pinned at `v4.31.0`.

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